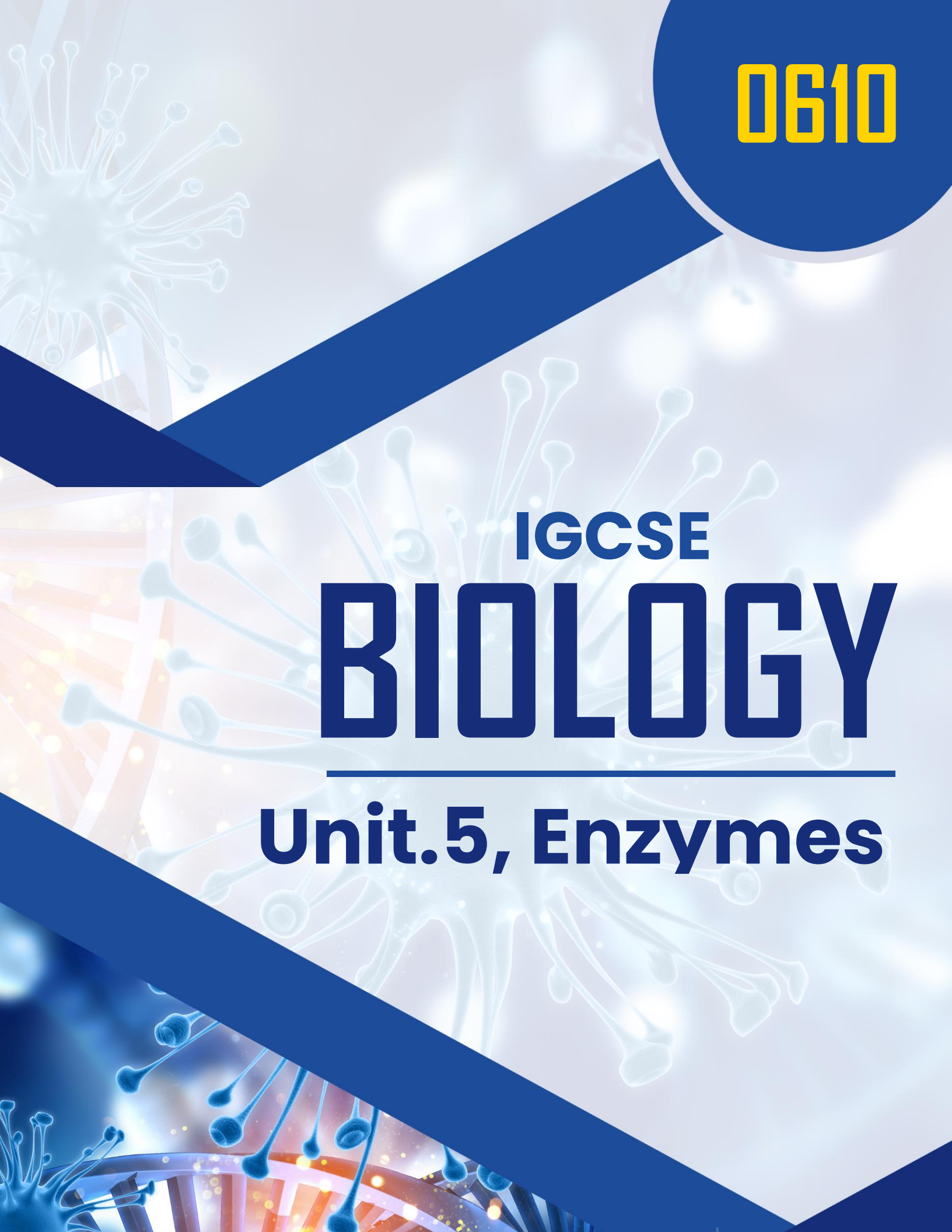




0610

IGCSE
BIOLOGY

Unit.5, Enzymes

Unit 5: Enzymes

Easy revision notes • diagrams • exam tips • practice questions with mark scheme

BY THE END OF THIS UNIT, YOU SHOULD BE ABLE TO:

- Say what an enzyme is and why living things need them
- Explain the lock and key model using the words **active site**, **substrate**, **product** and **enzyme-substrate complex**
- Describe and explain how **temperature** and **pH** change enzyme activity (including **denaturation**)
- Describe the amylase practical with iodine for temperature and for pH

5.1 What are enzymes?

- Enzymes are **biological catalysts**.
- A **catalyst** speeds up a chemical reaction and is **not used up or changed** at the end.
- **Biological** = made by living cells.
- All enzymes are **proteins**.
- Cells need enzymes so that **metabolic reactions** (all the chemical reactions that keep us alive) happen fast enough to support life.

Key word	Easy meaning
Catalyst	Something that speeds up a reaction and is not used up
Substrate	The molecule the enzyme works on (it gets broken down or joined together)
Active site	The special-shaped part of the enzyme where the substrate fits
Product	What is made in the reaction
Enzyme-substrate complex	Enzyme and substrate joined together while the reaction happens
Optimum	The best temperature or pH, where the enzyme works fastest
Denaturation	The enzyme loses its shape (active site changes) and stops working for good

STICKY NOTE – Remember it

- Enzyme = **protein + catalyst**. Reused again and again, never used up.

How do enzymes work?

Each enzyme has its own 3-D shape because it is a protein. The active site only fits **one** kind of substrate, the same way one key only fits one lock. This is why we say enzymes are **specific**.

Lock and key model

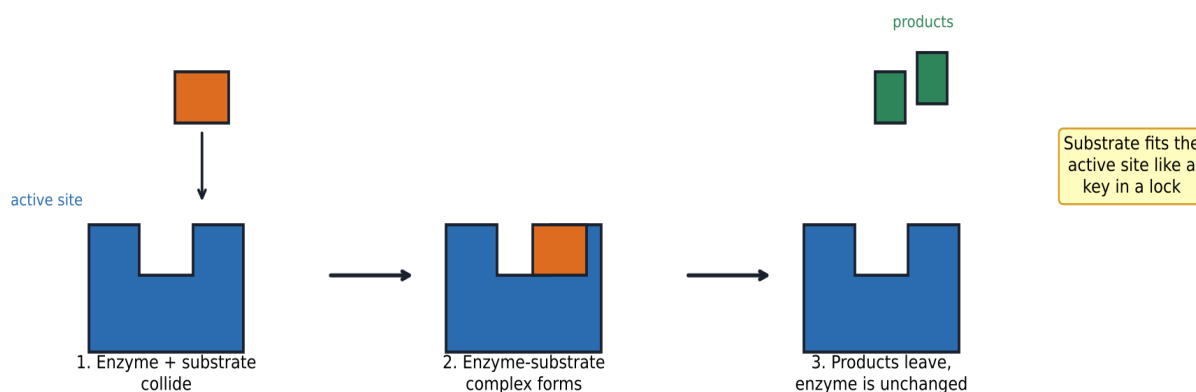


Figure 1: Lock and key model

Step by step:

1. Enzymes and substrates move around randomly in the solution.
2. They **collide**. If the substrate has the right shape it fits into the active site and an **enzyme-substrate complex** forms.
3. The reaction happens and the **products** are made. They no longer fit the active site so they leave.
4. The enzyme is **unchanged** and is free to take another substrate.

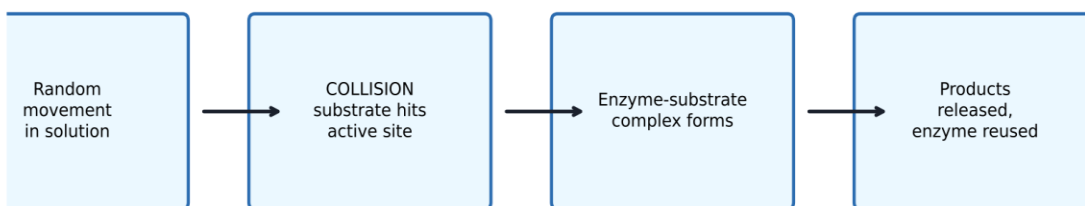


Figure 2: The 4 stages in one picture

EXAM TIP

- Use the exact words: "**substrate is complementary (matches) in shape to the active site**".
- Never say the enzyme "fits the substrate" or that the enzyme is "used up". That loses marks.

Enzymes and temperature

- Enzymes are proteins. Their shape is held by **bonds**. The shape of the **active site** is the most important part.
- Enzymes work fastest at their **optimum temperature**. For human enzymes this is about **37 °C**.
- Cold (0 °C up to the optimum)**: molecules have less energy, move slowly, so fewer collisions. The reaction is **slow** but the enzyme is **not damaged**.
- Too hot (above the optimum)**: the bonds break, the enzyme loses its shape, the active site changes. The substrate no longer fits. This is **denaturation**.
- Denaturation is **irreversible**. Cooling it down will not fix it.

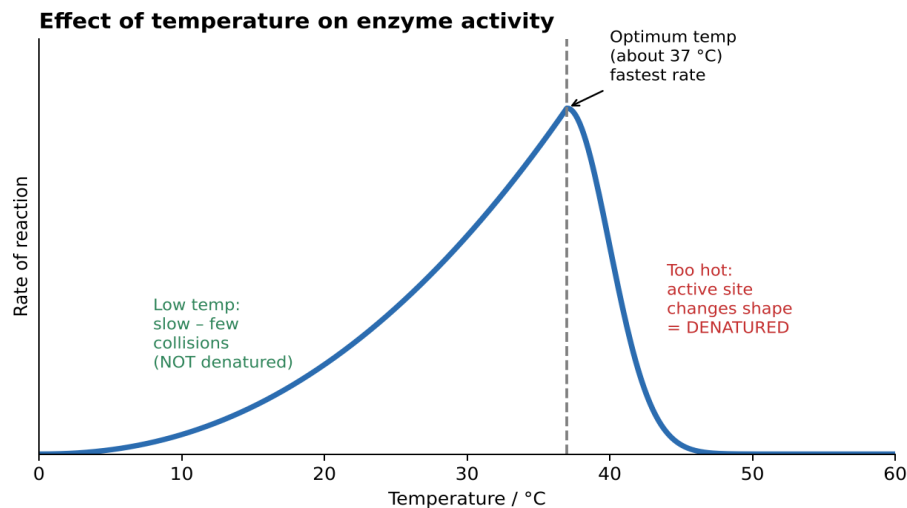


Figure 3: Rate against temperature

Denaturation

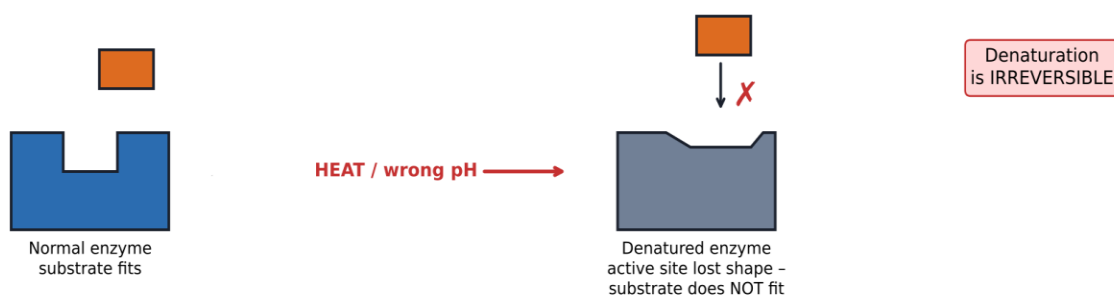


Figure 4: What denaturation does to the active site

STICKY NOTE – Cold vs hot

- Cold** = enzyme is **asleep** (slow, but alive). Warm it up and it works again.
- Hot** = enzyme is **broken** (denatured). It never comes back.

COMMON MISTAKE

- Do NOT write “the enzyme is killed”. Enzymes are not alive. Write **denatured**.
- Do NOT say low temperature denatures an enzyme. It only slows it down.

Enzymes and pH

- Most enzymes have an optimum of about **pH 7**.
- Some are different because of where they work:
 - **Stomach** enzymes (e.g. pepsin) like acid: about **pH 2**.
 - **Duodenum** enzymes like slightly alkaline: about **pH 8–9**.
- If the pH is too high or too low, the bonds holding the protein chain together are damaged. The **active site changes shape**, the substrate cannot fit, and the rate drops.
- Go far enough from the optimum and the enzyme is **denatured** and stops.

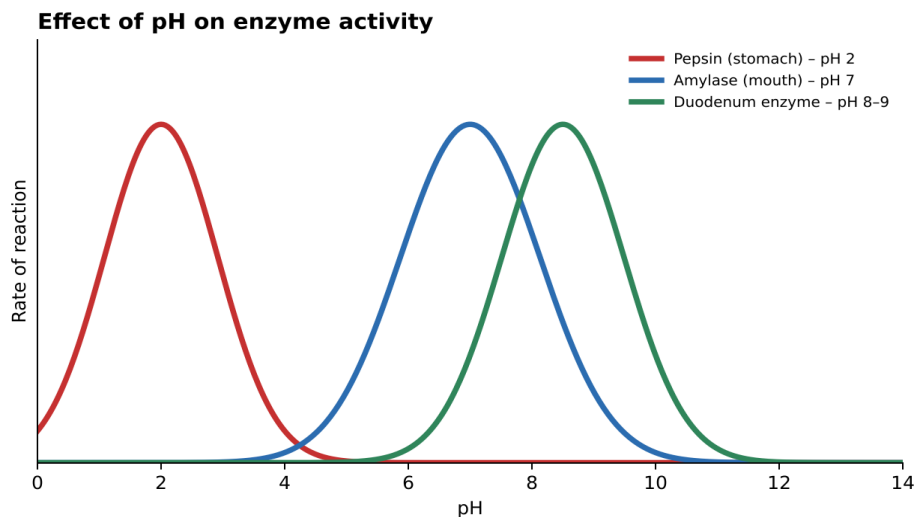


Figure 5: Each enzyme has its own optimum pH

STICKY NOTE – Quick link

- Temperature and pH have the **same effect** on the enzyme: **bonds break → active site changes → substrate does not fit → denatured**.

Temperature vs pH at a glance

	Too low / far below	Optimum	Too high / far above
Temperature	Slow rate, few collisions, NOT denatured	About 37 °C (fastest)	Denatured, activity stops
pH	Enzyme may denature (e.g. amylase at pH 2)	Depends on the enzyme (stomach 2, most 7, duodenum 8–9)	Enzyme may denature (e.g. pepsin at pH 9)

Practical 1: Effect of temperature on amylase

Aim: find how temperature changes the speed at which amylase digests starch.

1. Put drops of **iodine** in the wells of a spotting tile.
2. Heat the **starch solution** in a water bath to the temperature you are testing.
3. Add **amylase**, mix well and start the stopwatch.
4. Every minute take a drop of the mixture and add it to a new well of iodine.
5. Keep going until the iodine **stays orange-brown**. This means **no starch is left**.
6. Record the **time taken**. Repeat at other temperatures (e.g. 10, 20, 30, 40, 60 °C).

Starch-iodine test to follow amylase

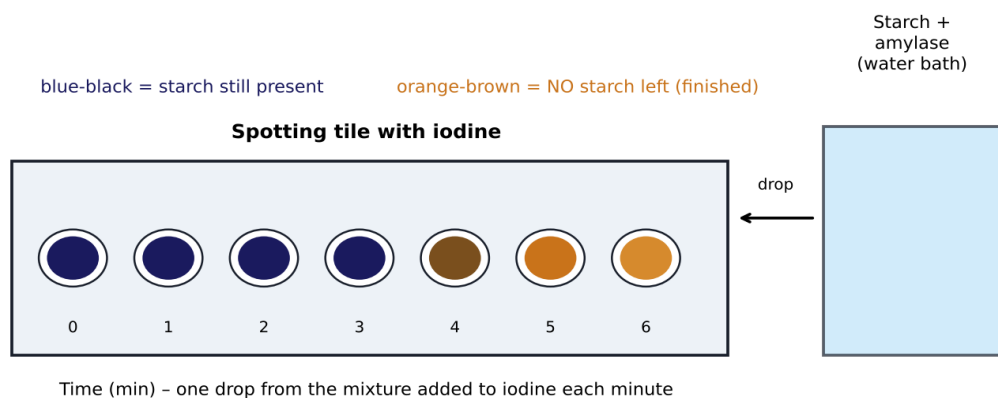


Figure 6: The spotting tile. Starch is gone when iodine stops turning blue-black

Result: the **shorter** the time, the **faster** the enzyme. The shortest time is at the optimum temperature.

Practical 2: Effect of pH on amylase

1. Put drops of **iodine** in rows on a spotting tile.
2. Label a test tube with the pH. Add **2 cm³ amylase** (syringe) and **1 cm³ buffer solution** (syringe).
3. In another tube measure **2 cm³ starch**. Add it to the amylase and buffer and **start the stopwatch**, mixing with a pipette.
4. After **10 seconds**, put one drop of the mixture onto the first drop of iodine (blue-black).
5. Repeat **every 10 seconds** until the iodine stays **orange-brown**.
6. Repeat using buffers of **different pH**. The shorter the time, the better the enzyme works at that pH.

EXAM TIP

- **Always name the colour change:** iodine goes from **orange-brown to blue-black** when starch is present.
- Variables to control: **volume and concentration** of enzyme and starch, **temperature** (for the pH test), **time between drops, same size drops**. Only change one thing.
- A buffer solution keeps the pH steady. If asked how to get different pH values, say **use buffer solutions**.
- To make results reliable: **repeat** and calculate a **mean**.

STICKY NOTE – Why a colour test?

- Iodine tests for **starch**. When amylase has digested all the starch, iodine **does not change colour**. No colour change = reaction finished.

Quick revision checklist

STICKY NOTE – Last-minute facts

- Enzyme = protein, biological catalyst, not used up.
- Active site matches substrate = **specific**.
- Optimum temperature (human) = **37 °C**. Stomach pH = **2**. Duodenum pH = **8–9**.
- Cold = slow. Hot or wrong pH = **denatured** (irreversible).
- Amylase digests **starch** (test: iodine, blue-black → orange-brown).

EXAM TIP

- “**Explain**” questions need a **because**. Use the chain: *energy → collisions → complexes* (cold side) or *bonds break → shape changes → no fit → denatured* (hot / pH side).
- On graph questions quote **numbers from the graph** with units, e.g. “rate rises from 2 to 12 units between 10 °C and 37 °C”.
- Look at the **marks**: 2 marks = 2 separate points.

Practice questions (exam style)

Total: **30 marks**. Try them without looking at your notes, then mark yourself with the mark scheme on the next pages. These are written in the style of IGCSE 0610 questions.

Question 1

(a) Enzymes are biological catalysts. State what is meant by the term **catalyst**. [2]

(b) Name the part of an enzyme where the substrate attaches. [1]

(c) Amylase can break down starch but it cannot break down protein. Explain why. [2]

Question 2

A student measured how fast an enzyme worked at different temperatures.

Temperature / °C	Rate of reaction / arbitrary units
10	2
20	5
30	9
37	12
45	6
55	0

(a) State the optimum temperature for this enzyme. [1]

(b) Describe and explain the change in rate between 10 °C and 37 °C. [3]

- (c) Explain why the rate is 0 at 55 °C. [3]

- (d) A second student says the enzyme would be denatured at 0 °C. Is she correct? Explain your answer. [2]

Question 3

A student investigated the effect of pH on the enzyme amylase. She mixed amylase, starch and a buffer solution. Every 10 seconds she tested a drop of the mixture with iodine solution.

pH of buffer	Time for iodine to stay orange-brown / s
2	Did not change in 300 s
5	90
7	30
9	150

- (a) State what iodine solution is used to test for and the colour change if it is present. [2]

- (b) Give **two** variables the student should keep the same. [2]

- (c) State how the student could make mixtures with different pH values. [1]

- (d) State the pH at which amylase works best. [1]

- (e) Explain the result at pH 2. [2]

Question 4

- (a) Pepsin is an enzyme in the stomach. Its optimum pH is 2. Predict and explain what happens to pepsin activity if it is moved into the duodenum, where the pH is 8. [3]

-
- (b) Explain why the rate of an enzyme-controlled reaction increases when the temperature rises from 20 °C to 37 °C. [2]

Question 5 (multiple choice)

- (a) Which statement about enzymes is correct? [1]
- A They are carbohydrates and are used up in the reaction
 - B They are proteins and are not changed by the reaction
 - C They are proteins and are used up in the reaction
 - D They are lipids and speed up reactions
- (b) Which change will denature an enzyme? [1]
- A Cooling to 0 °C
 - B Heating to 80 °C
 - C Adding more substrate
 - D Adding more water
- (c) In a starch and amylase investigation, which result shows that all the starch has been digested? [1]
- A Iodine stays orange-brown
 - B Iodine turns blue-black
 - C Iodine turns colourless
 - D Iodine turns green

Mark scheme

Written in the style of Cambridge mark schemes. A semicolon (;) separates marking points. A slash (/) means alternative wording is accepted. **max** = the most marks you can get for that part.

Q	Answer	Marks	Extra guidance
1(a)	speeds up (the rate of) a (chemical) reaction ; not used up / not changed (in the reaction) ;	2	accept “makes reactions faster”
1(b)	active site	1	do not accept “active place”
1(c)	active site of amylase is a specific shape / complementary to starch only ; protein does not fit the active site / no enzyme- substrate complex forms ;	2	ignore “amylase is the wrong enzyme” on its own
2(a)	37 (°C)	1	accept 37 ± 1 (units not needed)
2(b)	rate increases / gets faster (from 2 to 12) ; molecules gain (kinetic) energy / move faster ; more (frequent) collisions between enzyme and substrate / more enzyme-substrate complexes formed ;	3	ignore “more successful” without collisions
2(c)	high temperature breaks bonds (in the enzyme) ; shape of active site changes / lost ; substrate no longer fits (so no complexes form) / enzyme is denatured ;	max 3	do not accept “enzyme is killed”
2(d)	no / not correct ; low temperature does not denature enzymes / only slows the reaction down ; (because) less energy so fewer collisions ;	max 2	accept “enzyme works again when warmed”
3(a)	tests for starch ; (orange-brown to) blue-black ;	2	accept “black” or “dark blue”; do not accept “purple”
3(b)	any two from: volume of amylase / starch / buffer ; concentration of amylase / starch ; temperature ; size of drops / time between drops ;	2	do not accept “amount” on its own; do not accept pH

Q	Answer	Marks	Extra guidance
3(c)	use buffer solutions (of different pH)	1	accept “add acid and alkali” with a pH meter / indicator
3(d)	pH 7	1	
3(e)	pH 2 too acidic / too far from the optimum ; bonds break / active site changes shape / enzyme denatured ; starch is not digested / starch still present ;	max 2	ORA for other pH values
4(a)	activity decreases / stops ; pH 8 too alkaline / not the optimum ; bonds break / active site changes shape ; pepsin is denatured / substrate no longer fits ;	max 3	
4(b)	molecules have more (kinetic) energy / move faster ; more (frequent) collisions between enzyme and substrate / more enzyme-substrate complexes ;	2	
5(a)	B	1	
5(b)	B	1	
5(c)	A	1	

EXAM TIP

- Mark yourself honestly. Tick each ; that you wrote in your own words.
- Lost marks? Check the **why**: most students describe but forget to explain (energy → collisions, or bonds → shape → fit).

Total: 30 marks.